

National University of Computer and Emerging Sciences, Lahore Campus

	Course Name:	Professional Practices	Course Code:	CS 4001
	Program:	BS (Computer Science)	Semester:	Spring 2026
	Instructor:	Shehryar Raza (VF)	Section	CS (6C)
	Exam Type:	Quiz 2 - Version A	Total Marks:	20

Student Name

Roll No.

Q1. Encircle the correct option from the given Questions: [8 marks]

Which of the following describes the most frequent legal defense strategy used by defendants when facing a patent infringement lawsuit?

- A. Demonstrating that the invention was created independently without knowledge (prior) of the existing patent
- B. Arguing that original patent has failed the novelty or non-obviousness test**
- C. Proving that patent resulted in no actual financial loss to patent holder
- D. Asserting that patent had expired after the standard 70-year term for inventors

Under the Uniform Trade Secrets Act (UTSA), what is the critical requirement for information to be legally recognized as a trade secret?

- A. The information must register with state level agency every 10 years to remain valid
- B. It must be stored in a digital format and encrypted with at least a 256-bit key
- C. Company must take active and documented steps to protect and maintain information secrecy**
- D. It must be novel enough to pass the same tests required for a US patent

A senior engineer at a large firm signs a non-compete agreement stating they cannot work for a direct competitor within a 100-mile radius for two years. If they move to California and join a rival firm, how might the state's legal framework affect the agreement?

- A. Agreement is federally protected under the Espionage Act and must be enforced by California courts
- B. California largely prohibits non-competes making it un-likely it can be enforced by the courts**
- C. Engineer will be allowed to work but pay a fee equal to two years of salary to the original firm
- D. California courts will reduce the miles down to 10, but keep the two year duration

What is the "Public Domain Myth" frequently encountered in plagiarism cases involving digital content?

- A. The misconception that anything patented for over 20 years automatically loses its copyright
- B. The idea that information that is a trade secret becomes public domain if an employee remembers it
- C. The belief that all electronic content on the internet is free to use without attribution**
- D. The assumption that summarizing a source in your own words removes the need for attribution or citation

An engineer develops a unique algorithm that optimizes server cooling. However, the algorithm is based entirely on a well-known mathematical formula applied to a standard physical environment. According to the USPTO's four tests, why might this patent application be rejected?

- A. It fails the statutory clause test
- B. It fails the non obviousness test**
- C. It fails the utility test

D. Mathematical formulas are given an exception with eligibility if used for cooling
A tech company develops software that controls a physical manufacturing process (like curing rubber using sensors and timing algorithms). They want to patent this system, even though it includes software. Which legal principle supports their ability to patent it?

- A. Only the hardware components (like sensors) can be patented
- B. Software tied to a physical process can be patented
- C. All software code is automatically patentable
- D. Software is protected only under copyright, not patents

Unlike patents, trade secret protection is lost if which of the following occurs?

- A. The information is ruled to be obvious by a court
- B. The 20 year term limit is reached
- C. A competitor independently discovers the same information
- D. The company fails to file a registration with the USPTO

A company reverse engineers a competitor's software to make its product compatible with that system. Which legal doctrine allows this?

- A. The First Sale Doctrine
- B. The Patent Exhaustion Doctrine
- C. The Work for Hire Doctrine
- D. The Fair Use Doctrine

Q2. Carefully read the Question before attempting:

[12 marks]

NeuroLink Dynamics Ltd. is developing NeuroWeave, a smart healthcare platform that combines wearable devices, machine learning models, and a mobile ecosystem to monitor neurological disorders in real time. As the product nears launch, the company must carefully evaluate legal risks and intellectual property (IP) strategies, especially concerning ownership, licensing, and protection of innovation. Product components include:

1. **Signal Processing Algorithm:** A novel algorithm that filters and enhances brainwave signals with 70% higher accuracy than existing methods. It is embedded within the device firmware.
2. **ML Training Framework:** Built using a permissive open-source library (MIT License), but extended with proprietary optimization techniques developed internally. Mobile App
3. **UI/UX Design:** Designed by a freelance designer who was paid per milestone, but no formal IP assignment clause was included in the contract.
4. **Brand Identity "NeuroWeave: Decode the Mind":** Actively used across app stores, packaging, and global marketing campaigns.
5. **Patient Dataset Pipelines:** Includes curated datasets and preprocessing pipelines derived from confidential hospital partnerships under strict data-sharing agreements.
6. **Diagnostic Reports Generator:** Automatically produces structured medical reports using predefined templates and generated insights.

For each component (1–6), determine the most suitable form of intellectual property protection (Copyright, Patent, Trade Secret, or Trademark). Select exactly one:

1.	Patent	4.	Trademark
2.	Trade Secret	5.	Trade Secret
3.	Copyright	6.	Copyright